

JAYANI SAMARAKOON

Undergraduate in Software Engineering

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PROFESSIONAL SUMMARY

Software Engineering undergraduate with hands-on experience in full-stack web development, Android application development, REST API development, database integration, and software development using Git and GitHub. Experienced in building individual and team-based software projects involving frontend development, backend development, CRUD operations, database management, testing, deployment, and SDLC practices. Familiar with Python, Java, Kotlin, JavaScript, React.js, FastAPI, Flask, SQL, Firebase, SQLite, Docker, and Git. Also experienced in AI/ML and MLOps, including machine learning model development, model deployment, and automated ML pipelines. Seeking a Software Engineer Internship to contribute to real-world software projects while developing professional engineering skills.

EDUCATIONAL QUALIFICATIONS

BSc (Hons) in Software engineering

2023-Present

Sri Lanka Technology Campus (SLTC)

G.C.E. Advanced Level

Year of 2022

K/MW Pinnwala central college

Technology stream: Engineering technology , Science for technology , Information Technology

SKILLS

TECHNICAL SKILLS

Programming	Python, Java, Kotlin, JavaScript, C, C++, SQL
Frontend	React.js, HTML, CSS
Backend	FastAPI, REST APIs
Mobile	Flutter, Android Studio
AI/Machine Learning	Scikit-learn, Pandas, NumPy ,PyTorch
Databases	MySQL, Firebase
UI/UX	Figma, Wireframing, Prototyping
Tools	Git, GitHub, VS Code, Jira ,Docker,MATLAB,Google Colab,Jupyter

SOFT SKILLS

Team Collaboration & Leadership

Works effectively in teams and takes the initiative to achieve project goals.

Project Management

Plans, organizes, and completes tasks efficiently to meet deadlines.

Learning Agility

Quickly learns new tools, technologies, and techniques to improve performance.

PROJECTS

DeepShield: Explainable Multimodal Framework for Trust Assessment and Evidence-Based Fake News Detection

(Ongoing FYP - Expected Completion: 2027)

- Developing a multimodal AI system using text, image, and metadata for fake news detection.
- Integrating RAG and Explainable AI techniques to provide evidence-based predictions and a Trust Score.
- Building REST APIs using FastAPI and a responsive frontend using React.js.
- **Technologies:** *Python, PyTorch, Transformers, BERT, CLIP, OpenCV, FAISS, FastAPI, React.js.*

Mobile Application - MealMate (Ongoing - Individual)

- Developed a user-friendly mobile application for browsing and ordering food.
- Implemented user authentication with login and registration functionality.
- Designed responsive and intuitive user interfaces to improve the overall user experience.
- **Technologies:** *Android , Java, XML, Android Studio*
- **Github :** github.com/-MealMate-

Mobile Application – LankaSmartMart (Individual)

- Developed an Android application with user-friendly UI/UX designed using Figma.
- Integrated Firebase/MySQL for data storage and retrieval.
- Collaborated using Git/GitHub and followed SDLC practices including development, testing, and deployment.
- **Technologies:** *Figma, Android Studio (Java/Kotlin), Firebase/MySQL, Git, GitHub*
- **Github:** [LankaSmartMart---Android](#)

Mobile Application - EduTrack (Individual)

- Developed an Android student management application using Kotlin and Jetpack Compose.
- Implemented CRUD operations, student search, and SQLite database integration.
- **Technologies:** *Kotlin, Jetpack Compose, SQLite, Android Studio, Git & GitHub*
- **Github:** <https://EduTrack App2>

Web application - Inventory Management & Stock Prediction System (Individual)

- Developed a full-stack inventory management system using Flask.
- Implemented CRUD operations, responsive dashboard, and SQLite database integration.
- **Technologies:** *Python, Flask, SQLite, HTML, CSS, JavaScript*
- **Github:** [Inventory-management-system](#)

Web Application - Synthrium (Individual)

- Developed a web application for fruit and vegetable detection using image upload and live camera input.
- Built FastAPI REST APIs and integrated a Roboflow computer vision model.
- Developed a responsive interface and configured Vercel deployment.
- **Technologies:** *Python ,FastAPI, Roboflow ,Computer Vision ,HTML,CSS,JavaScript*
- **Github :** [AI-fruit-vegetable-detection](#)

CardioAI – Heart Attack Risk Prediction System (Individual)

- Built an end-to-end ML prediction system using 1,000 records and 9 cardiovascular features.
- Compared 4 classification models and achieved 86.19% ROC-AUC and 66.67% recall.
- Developed a Streamlit dashboard for prediction and model evaluation.
- **Technologies :** *Python, Scikit-learn, Streamlit, Pandas*
- **Github :** [CardioAI-Heart-Attack-Risk-Prediction-](#)

Hospital Appointment No-Show Prediction Using Machine Learning (Group)

Responsibilities:

- Developed a machine learning solution to predict patient appointment no-shows using healthcare data.
- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA).
- Built and evaluated a Logistic Regression classification model.
- Conducted Feature Importance Analysis to improve model interpretability.
- Developed a Streamlit web application for real-time appointment no-show prediction.
- **Technologies:** *Python, Pandas, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook, Streamlit, Joblib*
- **Github :** [smartcare-ai-project](#)

SMS SPAM Detection system (Group)

Responsibilities:

- Developed and trained a Random Forest classifier for SMS spam detection.
- Built and optimized an LSTM deep learning model for sequential text classification.
- Performed feature engineering and hyperparameter tuning to improve model performance.
- Designed and developed a responsive web-based user interface for real-time SMS spam prediction.
- Compared machine learning and deep learning model performance using Accuracy, Precision, Recall, F1Score, and ROC-AUC.
- **Technologies:** *Python, Random Forest, LSTM, Flask/FastAPI , HTML, CSS, JavaScript, Scikit-learn, TensorFlow*
- **Github :** [NLP-SMS-Spam-Detection](#)

REFERENCES

Available Upon Request